

Olist E-commerce Performance Analysis

Data Cleaning & Preparation

We initiated the analysis by performing structured data cleaning and validation using core data integrity principles.

1. Entity Integrity (Primary Keys)

We verified that all primary keys, such as `order_id` in the Orders table and `customer_id` in the Customers table, are unique and non-null. Any anomalies or duplicates were identified and handled to ensure each record represents a distinct entity.

2. Referential Integrity

We validated relationships across tables by ensuring that all foreign keys correctly map to corresponding primary keys. For example, `customer_id` in the Orders table was cross-checked against the Customers table to identify and resolve any orphan records.

3. Data Type Validation

We reviewed and corrected data types across tables to ensure consistency and usability. Date fields were standardised into proper date time formats, and numeric fields such as price and freight values were validated as decimal types. For instance, date fields in the Reviews table initially stored as VARCHAR were converted into appropriate date formats.

4. Duplicate Detection

We performed duplicate checks across key tables such as Customers and Sellers to eliminate redundant records and maintain data accuracy.

5. Business Rule Consistency

We enforced logical business rules to ensure data reliability. Examples include: „Every order must have a valid purchase timestamp, Payment values must be greater than zero, Delivered orders must have a delivery date

Data Understanding & Modelling

Following data cleaning, we conducted a detailed exploration of each table to understand its structure, attributes, and business significance.

The primary objective was to establish relationships across tables and design a logical data model. We identified key joins between entities such as Customers, Orders, Order Items, Payments, and Reviews.

Based on this, we structured the dataset into an **order-centric analytical model**, where each order serves as the central unit of analysis. This enabled seamless integration of customer behaviour, transaction details, product information, and delivery performance into a unified view.

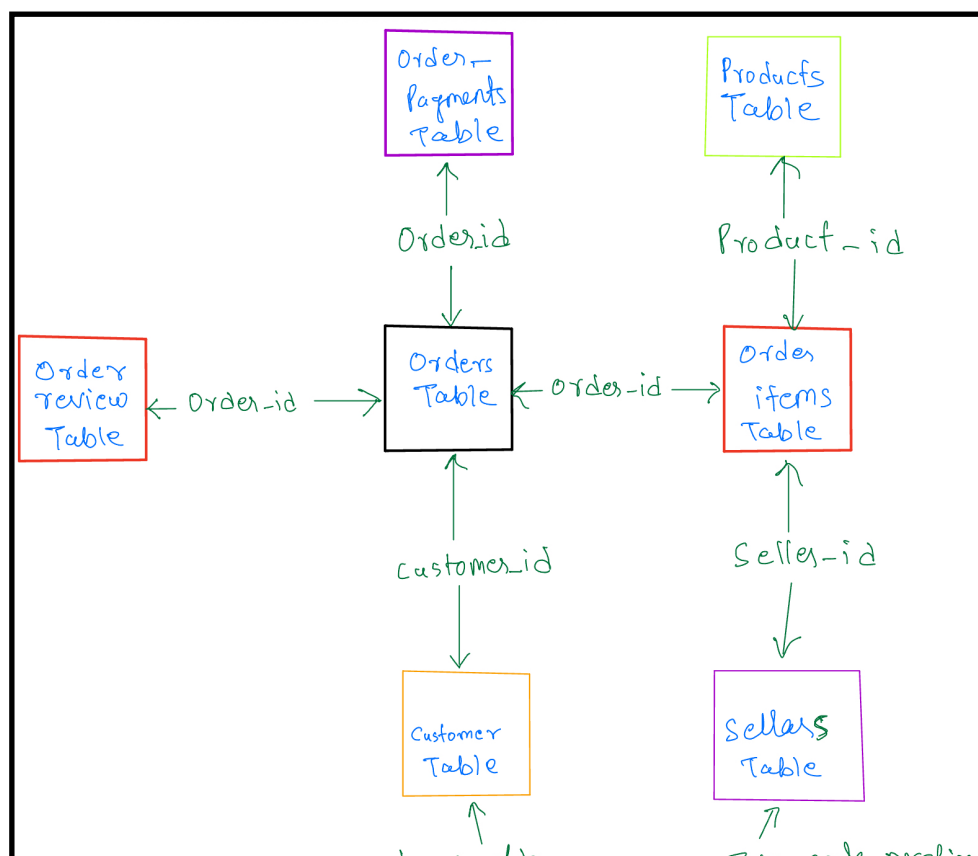
This data modelling approach ensured efficient querying, improved analytical clarity, and supported downstream funnel and performance analysis.

The table relationship can be understood from the diagram below:

Creation of Final Analytical Table :

We used the **Orders table as the base**, since it represents the core of all transactions. From there, we joined the **Customers, Order Payments, and Order Items** tables to bring in customer details, payment information, and product-level data.

Instead of joining everything at once, we kept the base table focused and only added other tables like **Products, Reviews, Sellers, or Geolocation** when needed for specific analyses. This helps keep the data easier to work with and avoids unnecessary complexity.



We observed that both the **Order Items** and **Order Payments** tables contain multiple records for the same `order_id`. This happens because a single order can include multiple products, and customers may also use multiple payment methods (e.g., instalments).

To avoid duplication and ensure accurate analysis, we aggregated these tables at the **order level**, summing up item prices and payment values. This allowed us to maintain a **single row per order** in the final dataset, ensuring consistency and preventing inflated metrics.

Data Analysis(Done with Advanced

SQL Querying in Dbeaver):

1. Funnel Conversion Optimisation

Table 1.1 : Funnel Conversion

Stage	Orders Count	Conversion %	Drop-off %
Order Created	99441	100.00%	—
Payment Completed	99437	99.99%	0.01%
Order Delivered	96478	97.02%	2.98%

Funnel Insight:

We analysed the customer journey across three key stages: order creation, payment completion, and delivery. The funnel highlights the conversion efficiency at each stage and identifies key drop-off points.

From the funnel analysis, we observe a **100% conversion rate from order creation to payment**, indicating that there are no significant issues in the payment process and customers are successfully completing transactions. However, a drop-off of approximately **2.98% occurs at the delivery stage**, suggesting that some orders are not being successfully fulfilled after payment.

This indicates a potential gap in **post-payment operations**, such as logistics, cancellations, or fulfilment issues. Further analysis is required at the **order status, regional, product category level** to identify the key drivers of this drop-off and quantify revenue leakage.

Recommendation:

Focus on reducing drop-offs at the delivery stage by improving logistics reliability and minimising cancellations. Strengthening post-payment operations can directly increase successful order completion. Even small improvements here can significantly boost overall conversion.

2. Status-wise Revenue Analysis

Table 2.1: Order_Status Vs Revenue

order_status	total_revenue	total_revenue_percent	Revenue Flag
delivered	1,54,22,462	96.34	Yes
invoiced	69,138	0.43	Not Yet
shipped	1,77,214	1.11	Not Yet
canceled	1,43,256	0.89	Never
unavailable	1,26,480	0.79	Never
processing	69,394	0.43	Not Yet
created	688	0	Not Yet
approved	241	0	Not Yet

Visualisation 2.1: Revenue status Exc Delivered Orders

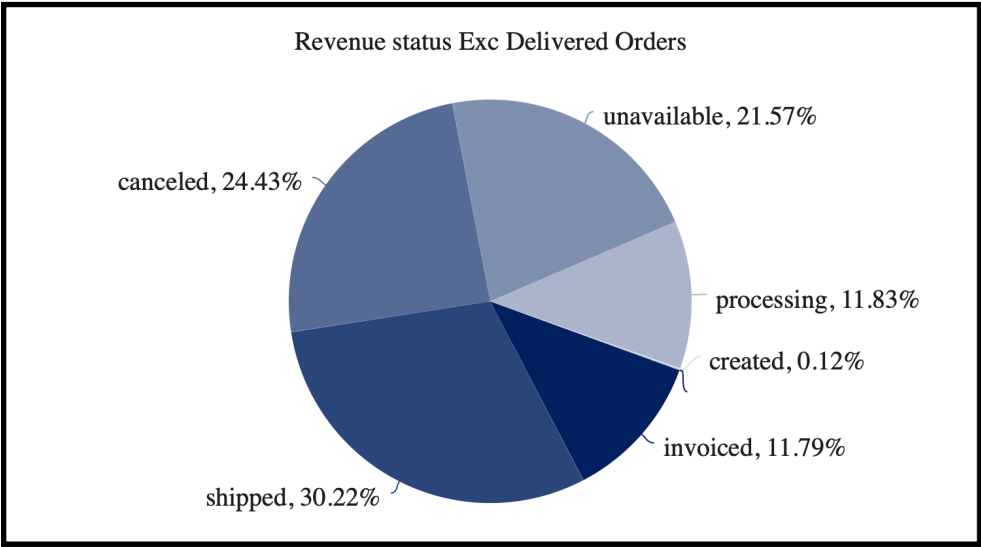


Table 2.1: Total_Revenue Vs Loss %

total_revenue(With all order status)	lost_revenue	loss_percentage
1,60,08,872	2,69,735.11	1.68

Insights:

From the status-wise revenue breakdown, we observe that orders marked as canceled and unavailable contribute the majority of revenue loss, accounting for approximately **1.68%** of total revenue. Other statuses such as **shipped, created,**

and approved represent orders that are still in progress and are expected to be completed. Therefore, they are not considered immediate contributors to revenue leakage.

Based on this, the primary area of concern is the **1.68% revenue loss driven by cancellations and unavailable orders**. These cases should be further analysed to identify underlying causes such as product availability issues, seller performance, or operational inefficiencies

Recommendation:

Prioritise reducing cancellations and unavailable orders by improving inventory accuracy and seller commitment. Implement stricter monitoring for order fulfilment and proactively address operational gaps causing revenue loss.

3. Status-wise Revenue Analysis

Table 3.1: Top 10 state by realised revenue:

customer_state	total_revenue	realized_revenue	revnue_pct
SP	59,98,226.96	57,70,266.19	37.41
RJ	21,44,379.69	20,55,690.45	13.33
MG	18,72,257.26	18,19,277.61	11.8
RS	8,90,898.54	8,61,802.4	5.59
PR	8,11,156.38	7,81,919.55	5.07
SC	6,23,086.43	5,95,208.4	3.86
BA	6,16,645.82	5,91,270.6	3.83
DF	3,55,141.08	3,46,146.17	2.24
GO	3,50,092.31	3,34,294.22	2.17

The revenue analysis shows a **high concentration of revenue in a few key states**, with **SP alone contributing ~37% of total revenue**, followed by **RJ (13%) and MG (11.8%)**. This indicates that a significant portion of the business is driven by a small number of regions. Additionally, the gap between total revenue and realised revenue is relatively small across all states, suggesting strong fulfilment efficiency and low revenue leakage in major markets.

However, since a large share of revenue is concentrated in a few regions, the business is **geographically dependent**, which could pose a risk. Expanding performance in mid-tier states like **RS, PR, and SC** presents an opportunity for growth.

Recommendation:

Target high-impact states like SP and RJ where volume is high and inefficiencies have larger impact. Improve logistics and seller performance in low-performing regions to reduce drop-offs and enhance delivery success rates.

4. State Wise Losses Analysis

Table 4.1: Top States by revenue with Drop Off rates:

customer_state	total_orders	delivered_orders	delivery_rate	dropped_orders	drop_off_pct
SP	41,746	40,501	97.02	1,245	2.98

MG	11,635	11,354	97.58	281	2.42
RJ	12,852	12,350	96.09	502	3.91
PR	5,045	4,923	97.58	122	2.42
SC	3,637	3,546	97.5	91	2.5

Table 4.2 : Top 5 States by revenue losses %:

customer_state	total_revenue	realized_revenue	lost_revenue	lost_revenue_pct
SP	59,98,226.96	57,70,266.19	1,19,290.97	1.99
MG	18,72,257.26	18,19,277.61	28,796.89	1.54
RJ	21,44,379.69	20,55,690.45	28,385.41	1.32
PR	8,11,156.38	7,81,919.55	16,503.18	2.03
SC	6,23,086.43	5,95,208.4	14,905.92	2.39

The top revenue-generating states show **high delivery success rates (>96%)**, indicating overall strong operational efficiency. However, despite this, there is still a **consistent drop-off of ~2–4% across all major states**, which translates into meaningful revenue loss due to their high order volumes.

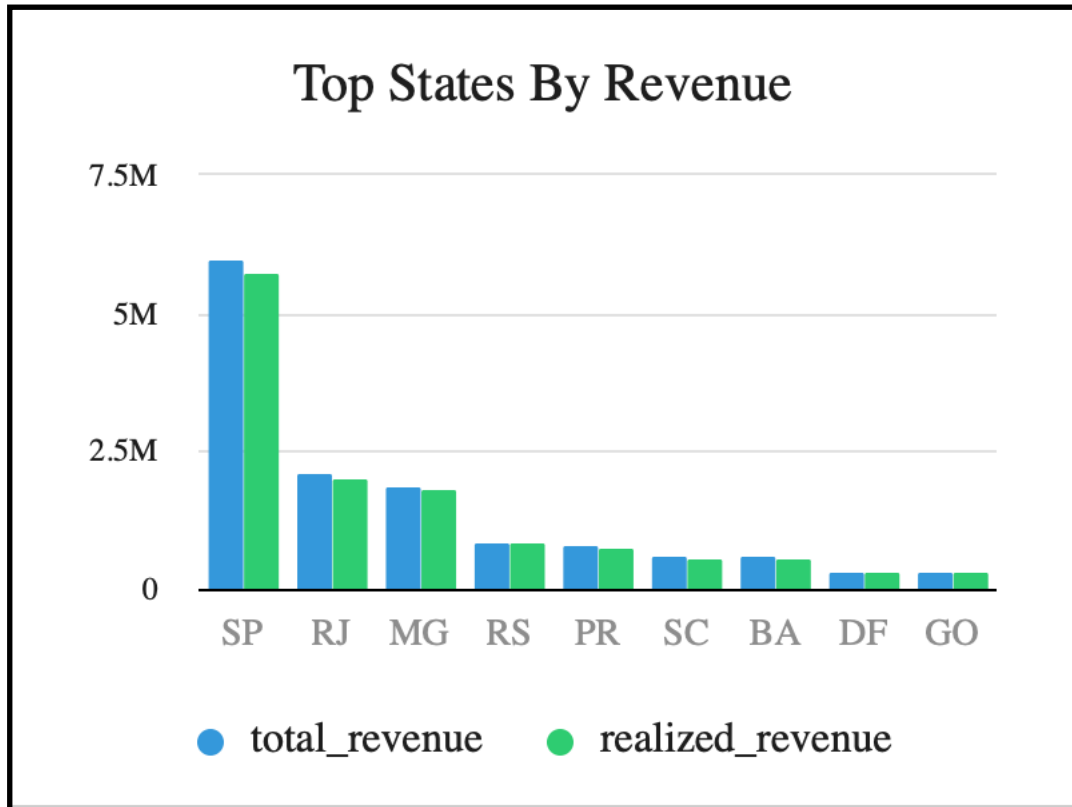
Among these:

- **RJ has the highest drop-off rate (~3.91%)**, indicating relatively weaker fulfilment performance
- SP contributes the highest absolute drop-offs (1,245 orders) due to its large volume
- Other states like **MG, PR, and SC perform better (~2.4–2.5%)**, but still present optimisation opportunities

This shows that even small inefficiencies in high-volume regions lead to **significant business impact**. We should start optimising operational efficiency in these states.

Table 4.3: Top 10 states by deliver drop off rates:

customer_state	total_orders	delivered_orders	delivery_success_rate
SE	350	335	95.71
CE	1,336	1,279	95.73
MA	747	717	95.98
RO	253	243	96.05
RJ	12,852	12,350	96.09
AL	413	397	96.13
PI	495	476	96.16
BA	3,380	3,256	96.33
PE	1,652	1,593	96.43
PB	536	517	96.46



5:

Product Category wise Revenue and loss Analysis:

Table 5.1: Top 5 product by realised revenue:

product_category_name_english	total_revenue	total_revenue_pct
health_beauty	12,58,681	9.26
watches_gifts	12,05,006	8.87
bed_bath_table	10,36,989	7.63
sports_leisure	9,88,049	7.27
computers_accessories	9,11,954	6.71

Table 5.2: Top 5 Product Category by revenue losses:

product_category_name_en	unavailable_orders	unavailability_pct	lost_revenue	lost_revenue_p
cool_stuff	16	0.44	20,053.15	2.57
garden_tools	13	0.37	14,762.95	1.76
computers_accessories	35	0.52	13,786.64	0.87
electronics	7	0.27	11,293.89	4.35
sports_leisure	47	0.61	10,764.33	0.77

The analysis shows that revenue is concentrated in a few key categories, with **health_beauty (9.26%)**, **watches_gifts (8.87%)**, and **bed_bath_table (7.63%)** being the top contributors. This indicates strong customer demand in lifestyle

and personal care segments. However, when analysing revenue losses, a different pattern emerges. Categories such as **cool_stuff**, **garden_tools**, and **electronics** contribute disproportionately to revenue leakage despite not being top revenue drivers.

Notably:

- **computers_accessories** and **sports_leisure** appear in both top revenue and loss categories
This indicates **high-volume categories with operational inefficiencies**.
- electronics shows a high loss percentage (4.35%) despite relatively low order volume. Suggests higher risk of cancellations or fulfilment issues

Recommendation:

Focus on categories contributing to both high revenue and high losses, such as computers_accessories and sports_leisure. Improve inventory planning and supplier reliability for high-risk categories like electronics to reduce cancellations.

6. Seller wise Delivery Revenue Analysis

Table 6.1: Top 10 Sellers with cancelled/unavailable orders(Minimum of 100 delivered orders)

seller_id	total_orders	unfulfilled_orders	unfulfilled_order_s_pct	total_revenue	lost_revenue	lost_revenue_pct
7e93a43ef30c4f03f38b393420bc753a	336	4	1.19	1,82,245.57	3,915.63	2.15
f5a590cf36251cf1162ea35bef76fe84	117	3	2.56	57,440.94	1,571.98	2.74
620c87c171fb2a6dd6e8bb4dec959fc6	740	8	1.08	1,28,919.29	1,234.93	0.96
fa1c13f2614d7b5c4749cbc52fecda94	585	1	0.17	2,03,811.38	1,223.06	0.6
36a968b544695394e4e9d7572688598f	154	1	0.65	11,691.73	1,161.47	9.93
46dc3b2cc0980fb8ec44634e21d2718e	521	3	0.58	1,39,567.05	989.91	0.71
17e34d8224d27a541263c4c64b11a56b	255	2	0.78	61,639.77	924.53	1.5
a416b6a846a11724393025641d4edd5e	162	5	3.09	28,670.52	804.72	2.81
cc419e0650a3c5ba77189a1882b7556a	1,706	9	0.53	1,30,557.55	787.46	0.6
b2ba3715d723d245138f291a6fe42594	337	4	1.19	58,128.53	763.18	1.31

The analysis of top sellers with canceled or unavailable orders shows that **revenue leakage is concentrated among a small group of sellers**, though the overall percentage of unfulfilled orders remains relatively low across most sellers. Most high-volume sellers (e.g., seller with 1,706 orders) maintain **low unfulfilled rates (~0.5%)**, indicating generally strong performance. This indicates that a small number of underperforming sellers are driving a significant portion of revenue leakage

Recommendation:

Identify and closely monitor underperforming sellers with high cancellation or loss rates. Implement seller performance benchmarks and incentives reliable sellers while limiting exposure of consistently poor performers.

7. Delivery Time Analysis:

Table 7.1: Average Delivery time in days:

total_delivered_orders	96,478
avg_delivery_days	12.5
fastest_delivery	0
slowest_delivery	210

The analysis shows that the **average delivery time is approximately 12.5 days** across ~96,000 delivered orders, indicating a moderately long fulfilment cycle.

However, there is **significant variability in delivery performance**:

- Fastest delivery: same-day (0 days)
- Slowest delivery: 210 days

This wide range highlights **inconsistency in logistics and fulfilment operations**

8. Correlation between Delivery time and review score:

Table 8.1 : Review Vs Average delivery days

review_score	avg_delivery_days	total_orders
1	21.21	7,417
2	16.36	2,311
3	14.21	6,243
4	12.23	14,960
5	10.6	44,756

Table 8.2 : Review Vs Delivery Bucket

delivery_bucket	avg_review_score	total_orders
Very Fast (0-3 days)	4.46	5,478
Fast (4-7 days)	4.39	18,721
Moderate (8-10 days)	4.33	15,104
Slow (11-15 days)	4.25	17,310
Very Late (16+ days)	3.6	19,074

The analysis shows a **clear negative relationship between delivery time and customer satisfaction**.

From Table 8.1: Orders with 1-star reviews take ~21 days on average while orders with 5-star reviews are delivered in ~10.6 days.

This shows that **faster deliveries consistently lead to higher customer satisfaction**

From Table 8.2:

- Very Fast deliveries (0–3 days) → Avg rating 4.46
- Very Late deliveries (16+ days) → Avg rating 3.6

A drop of nearly **0.8 rating points**, which is significant at scale

Customer satisfaction declines significantly as delivery time increases, with delayed deliveries (16+ days) receiving substantially lower ratings, highlighting delivery speed as a key driver of customer experience.

Recommendation:

Prioritise faster deliveries, especially keeping delivery time within 10 days, to maintain high customer satisfaction. Focus on reducing long-tail delays (15+ days), as they have the strongest negative impact on reviews.

9. Customer Satisfaction Analysis:

Table 9.1: Most Dissatisfied Customer segment by state:

customer_state	avg_review_score	total_orders
AL	3.73	328
MA	3.79	602
SE	3.82	273
PA	3.84	763
BA	3.85	2,672
CE	3.87	1,024
RJ	3.88	10,007
PI	3.98	382
PE	3.99	1,278
GO	4.02	1,577

Table 9.2: Review Score by customer order value:

customer_segment	avg_review_score	total_orders
High Value	3.88	3,364
Medium Value	4.04	37,641
Low Value	4.15	36,912

1. Regional Dissatisfaction

The analysis shows that certain states consistently report **lower average review scores**, with:

- **AL (3.73), MA (3.79), SE (3.82)** among the lowest
- Followed by **BA and RJ**, which are also high-volume states

This indicates that **customer experience is weaker in specific regions**, likely due to:

- Longer delivery times
- Logistics inefficiencies
- Seller performance issues

2. Customer Value-Based Dissatisfaction

A clear trend emerges across customer segments:

- **High-value customers** → **lowest satisfaction (3.88)**
- **Medium-value customers** → **moderate satisfaction (4.04)**
- **Low-value customers** → **highest satisfaction (4.15)**

This suggests that:

- High-value customers have **higher expectations**
- They are more sensitive to:
 - Delivery delays
 - product/service quality

Recommendation:

Improve service quality in low-performing regions and prioritise high-value customers who show lower satisfaction. Providing faster delivery and better support for premium customers can improve retention and lifetime value.

10. Repeat Customer and Customer Retention:

Visualisation 10.1: Repeat Customers:

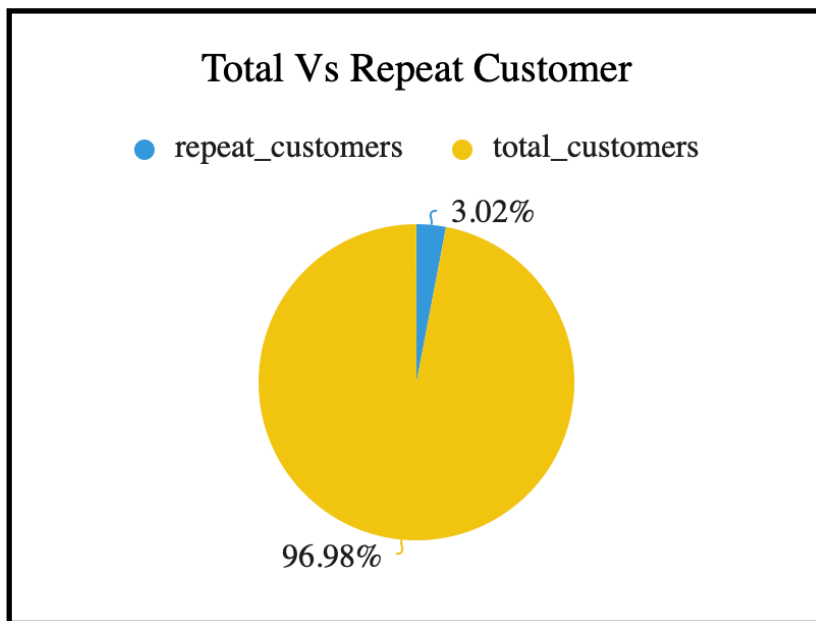


Table 10.1 : Repeat vs New Customer Review:

customer_type	avg_review_score	total_orders
---------------	------------------	--------------

New Customer	4.08	72,556
Repeat Customer	4.12	5,361

Customer Retention & Satisfaction

The analysis shows that only **~3.12% of customers are repeat buyers (2,997 out of 96,096)**, indicating a **very low customer retention rate**. This suggests that the majority of customers are one-time buyers, which points to a potential gap in long-term customer engagement and loyalty.

Note: It is because the data which we are analysing is a snapshot of a large dataset.

Customer Experience Perspective

Interestingly, repeat customers have a **slightly higher average review score (4.12)** compared to new customers (4.08). This indicates:

- Customers who return generally have a **better experience**
- The issue is not satisfaction alone, but **lack of retention mechanisms**

Recommendation:

Introduce retention strategies such as loyalty programs, personalised offers, and post-purchase engagement to convert one-time buyers into repeat customers. Leveraging already high satisfaction levels can significantly improve repeat purchase rates.

Conclusion:

This analysis highlights that while the e-commerce platform demonstrates strong performance in terms of **payment conversion and overall delivery success**, there are critical inefficiencies within the **post-payment and fulfilment stages**.

A consistent drop-off of ~3% during delivery, along with revenue leakage from cancellations and unavailable orders, indicates gaps in **logistics, inventory management, and seller reliability**. Additionally, delivery delays have a direct negative impact on customer satisfaction, with longer delivery times significantly reducing review scores.

The analysis also reveals that **revenue is concentrated in a few key regions and product categories**, making operational inefficiencies in these areas more impactful. At the same time, **customer retention remains very low (~3%)**, despite relatively high satisfaction levels, suggesting missed opportunities in converting one-time buyers into loyal customers.

Overall, the key opportunities lie in:

- Improving **delivery speed and consistency**
- Reducing **seller-driven cancellations and stock issues**
- Strengthening **customer retention strategies**

Addressing these areas can significantly enhance **customer experience, operational efficiency, and long-term business growth**.